

SkillOps: A Practical Framework for Designing, Testing, and Operating Modular Skills in Personal AI Agents

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Abstract

Personal AI agents increasingly rely on reusable capability modules, often called skills, yet their design and operation remain largely informal. This paper introduces SkillOps, a practical framework for designing, testing, and operating modular skills in personal AI agents. The framework is derived from open-source artifacts developed by the author and contributes a skill component taxonomy, a life-cycle model, a failure-mode taxonomy, and an artifact-based evaluation design. The paper reports descriptive benchmark summaries from a manually constructed exploratory benchmark covering artifact structure, trigger-routing cases, and operational-risk cases, together with a controlled repeated pilot on `deepseek/deepseek-chat` preserved in the repository. These materials provide preliminary observations only and do not claim statistical significance, broad empirical validation, or a general multi-provider comparison. The contribution is therefore a modest engineering framework for making agent skills more explicit, testable, and maintainable.

1 Introduction

Large language model agents are increasingly used as interactive systems that can read instructions, call tools, edit files, use external APIs, and maintain task-specific context. In such systems, user-visible behavior is shaped not only by the base model, but also by auxiliary capability modules. These modules may be called tools, plugins, agents, memories, workflows, or skills. This paper focuses on the skill as a practical unit of agent capability in personal agent environments.

In many personal agent settings, skills are written and maintained by individual users or small teams rather than by centralized platform engineers. A skill may contain a trigger description, operational instructions, examples, constraints, memory rules, safety checks, and tests. When these elements are implicit or poorly specified, the agent may activate the wrong skill, inject excessive context, ignore constraints, retain obsolete information, or perform unsafe operations. These are not only model-quality issues; they are operational issues in the design, testing, and maintenance of modular agent capabilities.

This paper introduces SkillOps, a practical framework for the lifecycle management of agent skills. The term is used by analogy to operational practice in software engineering, but the scope is narrower. SkillOps does not propose a new model architecture or a general theory of agents. Instead, it provides a structured way to define, test, audit, and revise skills as reusable capability units.

The framework is derived from five open-source artifacts:

- `skill-design-guide`: guidance for defining skill structure and usage boundaries.
- `skill-security-guard`: checks for unsafe or ambiguous skill behavior.

- `persistent-memory`: a file-based memory mechanism for continuity across agent sessions.
- `agent-self-audit`: a self-auditing mechanism for reviewing agent outputs and operational risks.
- `lobster-guard`: a guardrail-oriented tool for checking operational assumptions and failure modes.

The central claim is modest: personal agent skills become easier to inspect, test, and govern when they are treated as lifecycle-managed artifacts with explicit structure, tests, and operational controls. The paper contributes a framework, a taxonomy of skill components, a methodology for artifact-based study, a descriptive benchmark layer built from manually constructed inputs, and controlled repeated pilot observations preserved in the repository. The reported tables and pilot summaries should not be read as broad empirical validation or as a general multi-provider comparison.

2 Background and Motivation

2.1 From Prompts to Operational Skills

Prompt engineering is often discussed as the design of instructions for a language model. In personal agent systems, however, instructions are rarely isolated. They may be tied to tools, files, context windows, memory policies, or task-specific workflows. A skill can therefore be understood as a structured capability unit that combines instructions with operational assumptions.

For example, a writing skill may define when it should be used, what tone it should adopt, what sources it may rely on, and how it should handle uncertainty. A code-review skill may specify repository access assumptions, test execution rules, and failure-reporting requirements. A memory skill may specify what information should be retained, where it should be stored, and how obsolete information should be forgotten. These examples suggest that a skill is closer to an operational module than to a single prompt.

2.2 Operational Failure Modes

Informal skills can fail in several recurring ways. A skill may be triggered too broadly or too narrowly. It may inject irrelevant or stale context, causing the model to overfit to past assumptions. It may contain constraints that are underspecified or not testable. It may introduce safety risks when it authorizes file edits, command execution, credential handling, or external communication. It may also accumulate obsolete memory that conflicts with current user intent.

These failure modes motivate a more systematic approach. If skills are treated as artifacts, they can be linted, tested, reviewed, versioned, and retired. This does not eliminate model uncertainty, but it can make operational assumptions more visible and auditable. Table 1 summarizes the recurring failure categories used throughout the framework.

2.3 Positioning of SkillOps

SkillOps is positioned between prompt design, tool-use agents, and software engineering practice. It does not attempt to replace agent architectures such as tool-use planning, reasoning-action loops, or autonomous software engineering agents. Instead, it addresses a complementary layer: how reusable skills should be specified and operated in a personal agent environment.

Prior work on tool-using agents studies how language models decide when and how to use external tools [23, 32]. Work on agent-computer interfaces studies how environment design affects agent performance [31]. Work on reusable skill libraries shows that agent capabilities can be stored and reused over time [30]. SkillOps builds on this general direction but focuses on the practical lifecycle of user-defined skills: design, triggering, context injection, constraint enforcement, auditing, and forgetting.

Failure mode	Description	Typical mitigation
Over-triggering	The skill activates for requests that only superficially match its scope.	Narrower trigger language, explicit negative examples, boundary tests.
Under-triggering	The skill fails to activate for valid requests.	Positive examples, paraphrase coverage, and missed-activation checks.
Context contamination	Irrelevant, stale, or conflicting context is injected into execution.	Source filtering, freshness checks, conflict-resolution rules.
Constraint drift	The skill’s constraints are vague, contradictory, or not enforced consistently.	Testable rules, lint checks, regression cases.
Unsafe tool use	The skill authorizes risky file, command, network, or credential operations without sufficient guardrails.	Permission scoping, security scanning, post-run audit questions.
Memory drift	Stored memory persists after the underlying assumption has changed.	Expiry rules, deletion paths, explicit retirement and forgetting policies.
Audit blindness	Failures are not surfaced because the skill does not require self-review or evidence reporting.	Output contracts, self-audit prompts, artifact logging.

Table 1: Failure-mode taxonomy for skill-based personal agents.

3 The SkillOps Framework

3.1 Definition of a Skill

In this paper, a skill is defined as a reusable capability unit that guides an agent’s behavior for a class of tasks. A skill may include natural-language instructions, examples, trigger conditions, context requirements, tool permissions, safety constraints, output contracts, and tests.

SkillOps-managed skills are intended to document these assumptions in a more structured and reviewable form. Table 2 summarizes the structural components used in the framework, and Figure 1 provides a schematic view of how those components fit together in a skill package.

3.2 Lifecycle Model

SkillOps organizes skill management into a lifecycle:

1. **Design:** define purpose, scope, triggers, constraints, and output expectations.
2. **Lint:** check for ambiguity, missing fields, conflicting instructions, unsafe permissions, and untestable claims.
3. **Test:** evaluate the skill on positive, negative, boundary, and adversarial examples.
4. **Deploy:** make the skill available to the agent under controlled activation conditions.
5. **Observe:** collect failures, unexpected activations, user corrections, and context conflicts.
6. **Audit:** review outputs and operational traces for risk, drift, and non-compliance.
7. **Revise:** update the skill based on observed failures.
8. **Forget or retire:** remove obsolete context, deprecated rules, or unsafe skills.

The lifecycle emphasizes that a skill is not complete when it is written. Its behavior depends on activation, context, model behavior, and interaction with other skills. Figure 2 summarizes the lifecycle stages from design through forgetting or retirement.

Component	Operational role	Example question
Name and purpose	Defines the task class and intended value of the skill.	What problem is this skill meant to solve?
Trigger boundary	States when the skill should and should not activate.	What requests are in scope, out of scope, or ambiguous?
Input assumptions	Records required user or environment information.	What must be known before execution starts?
Context policy	Specifies what prior context, files, or memory may be injected.	What supporting context is necessary, and how fresh must it be?
Execution constraints	States required, optional, and forbidden actions.	What must the agent do, avoid, or verify?
Tool and file permissions	Limits resource access during execution.	Which tools, files, networks, or services are allowed?
Output contract	Defines the expected form of the answer or artifact.	What must the final output contain?
Safety and security checks	Surfaces risks and policy-relevant conditions.	What unsafe actions, hidden assumptions, or trust-boundary issues must be checked?
Tests and examples	Provides positive, negative, and edge cases.	How is the skill regression-tested?
Lifecycle metadata	Records versioning, revision triggers, and retirement rules.	When should the skill be updated, deprecated, or forgotten?

Table 2: Structural components of a SkillOps-managed skill.

3.3 Trigger Design

Trigger descriptions determine when a skill is activated. A broad trigger may widen activation coverage but also cause irrelevant activation. A narrow trigger may reduce false positives but fail to activate in valid cases. SkillOps therefore recommends that trigger descriptions include both positive and negative examples.

A trigger specification should answer three questions:

- What user requests should activate the skill?
- What superficially similar requests should not activate it?
- What ambiguity should cause the agent to ask for clarification or fall back to a general response?

Trigger quality can be evaluated through paraphrase tests, boundary tests, and multi-intent tests. This paper reports only the descriptive composition of a manually constructed trigger benchmark; Section 6 summarizes the reported counts and their limitations.

3.4 Context Injection

Context injection is necessary when a skill depends on project conventions, user preferences, prior decisions, or repository structure. However, excessive context can degrade behavior by introducing stale assumptions or irrelevant constraints. SkillOps treats context injection as a declared policy rather than opportunistic accumulation of memory.

A context injection policy should specify:

- the minimum context needed for the skill to operate;
- the source of the context, such as files, memory entries, or tool outputs;
- the freshness requirements for the context;

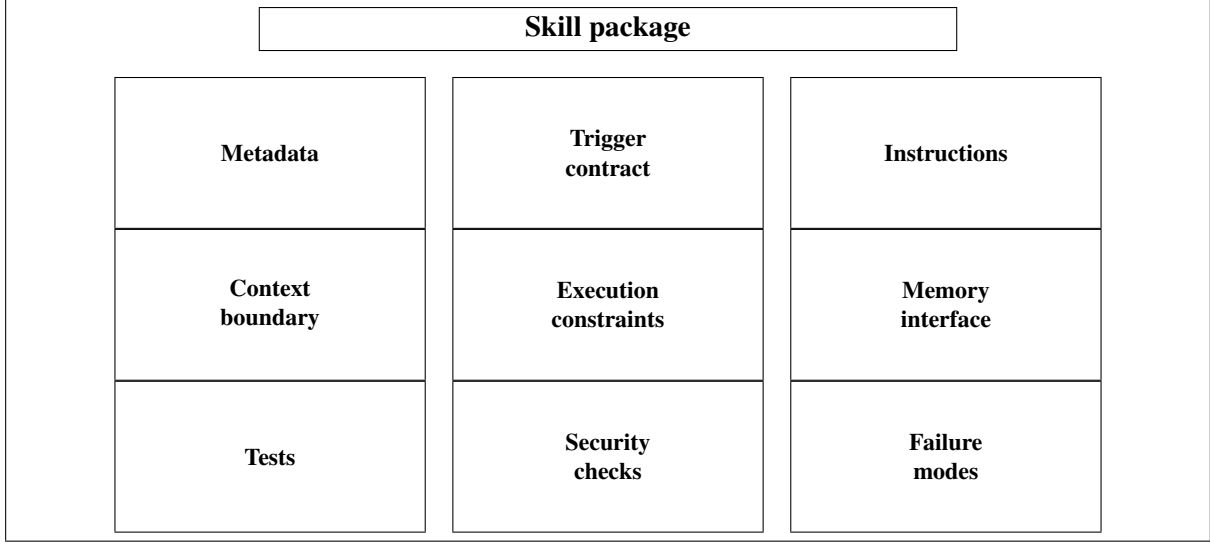


Figure 1: LaTeX-native rendering of a SkillOps skill package, with the package boundary enclosing meta-data, trigger contract, instructions, context boundary, execution constraints, memory interface, tests, security checks, and failure modes.

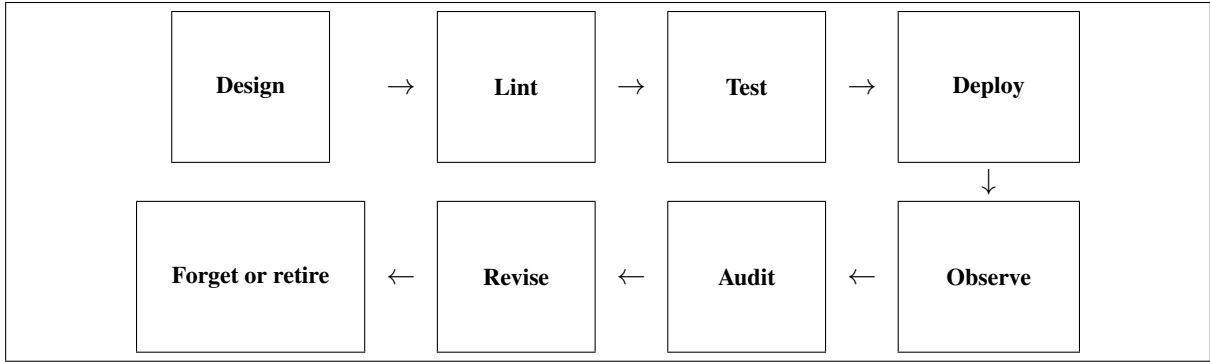


Figure 2: LaTeX-native rendering of the SkillOps lifecycle, showing the linked sequence of design, lint, test, deploy, observe, audit, revise, and forgetting or retirement.

- the conditions under which context should be omitted;
- the method for resolving conflicts between current user instructions and stored context.

This is especially important for personal agents, where long-lived memory can improve continuity but also preserve outdated assumptions.

3.5 Execution Constraints

Execution constraints define what the agent is allowed to do during skill use. For low-risk writing skills, constraints may concern tone, citation style, or formatting. For higher-risk skills, constraints may concern file modification, command execution, external communication, credential handling, or irreversible actions.

SkillOps distinguishes between three types of constraints:

- **Behavioral constraints:** rules about tone, reasoning style, response structure, or uncertainty reporting.

Dimension	Informal skill	SkillOps-managed skill
Scope definition	Implicit task description embedded in prompt text.	Named purpose, explicit scope, and documented non-scope.
Activation	Heuristic or ad hoc triggering.	Trigger contract with positive, negative, and boundary examples.
Context handling	Opportunistic context accumulation.	Declared context sources, freshness rules, and conflict handling.
Constraints	Broad instructions such as “be careful.”	Testable behavioral, operational, and safety constraints.
Testing	Rare or manual spot checks.	Regression cases, adversarial examples, and lint coverage.
Security review	Often absent or implicit.	Permission scoping, unsafe-action checks, and trust-boundary review.
Memory handling	Persistent context without explicit expiry rules.	Defined write permissions, forgetting policy, and retirement criteria.
Operational reporting	Minimal trace of what changed and why.	Output contract, audit fields, versioning, and revision history.

Table 3: Comparison between informal prompt-based skills and SkillOps-managed skills.

- **Operational constraints:** rules about tools, files, commands, commits, network access, and external services.
- **Safety constraints:** rules about privacy, credentials, harmful actions, or irreversible operations.

A constraint is more useful when it is testable. For example, “do not claim success without evidence” is more testable than “be careful.” SkillOps therefore favors constraints that can be checked through examples, lint rules, or audit questions.

3.6 Forgetting and Retirement

Forgetting is a necessary part of skill operation. A personal agent may retain preferences, project state, or procedural rules that later become obsolete. Without a forgetting mechanism, the agent may apply outdated instructions to new tasks.

SkillOps treats forgetting at two levels. First, individual memory entries may need to be removed or revised. Second, entire skills may need to be deprecated when their scope changes or when they become unsafe. Retirement rules should specify how to identify obsolete skills, how to preserve historical context if needed, and how to prevent deprecated instructions from continuing to influence behavior.

Table 3 contrasts informal prompt-based skills with skills managed under the proposed framework.

4 Research Questions

This paper is organized around three research questions.

RQ1: What structural components should a skill include as an agent capability unit? This question examines whether a practical skill template can make agent capabilities easier to understand, test, and maintain. The expected output is a component taxonomy covering triggers, context, constraints, tool access, output contracts, tests, and lifecycle metadata.

Research question	Framework component	Evaluation method
RQ1	Skill structure and documentation fields	Artifact review, template comparison, and completeness analysis across the open-source repositories.
RQ2	Trigger design, context policy, execution constraints, and forgetting	Manually constructed benchmark cases covering activation boundaries, stale-context injections, and deprecated-memory scenarios.
RQ3	Linting, security scanning, and self-audit	Seeded failure cases, reviewable repository checks, and manual review criteria for operational risk inspection.

Table 4: Mapping from research questions to framework components and evaluation methods.

Artifact	Role in SkillOps
<code>skill-design-guide</code>	Provides design guidance for structuring skills, defining scope, and writing trigger descriptions.
<code>skill-security-guard</code>	Provides checks for unsafe, ambiguous, or overly permissive skill behavior.
<code>persistent-memory</code>	Provides a file-based approach to retaining and distilling long-term agent context.
<code>agent-self-audit</code>	Provides a mechanism for reviewing agent outputs, assumptions, and possible failure modes.
<code>lobster-guard</code>	Provides operational guardrails for checking assumptions, reporting partial failure, and avoiding unsupported claims.

Table 5: Open-source artifacts used as the practical basis for the SkillOps framework.

RQ2: How do trigger descriptions, context injection, execution constraints, and forgetting affect agent stability? This question examines how design choices influence activation behavior, consistency across paraphrases, constraint following, and resistance to stale context. The expected output is a benchmark suite with controlled variations of skill definitions.

RQ3: How can linting, security scanning, and self-auditing support operational risk review through reviewable checks? This question examines how reviewable checks can surface common skill risks and make them easier to inspect. The expected output is a manually constructed set of lint, security, and self-audit cases together with repository-level checks and review criteria for operational risk review. It does not test whether these mechanisms reduce deployment risk or improve model behavior.

Table 4 maps each research question to the corresponding framework component and evaluation method.

5 Artifacts and Methodology

5.1 Artifact Base

The framework is derived from a set of open-source artifacts. Table 5 summarizes their intended roles.

The repository location for the paper and its supporting materials is provided in Section 5.4.

5.2 Methodological Approach

This paper uses an artifact-based methodology. Rather than beginning with a large-scale user study, it derives a framework from the design and use of practical open-source tools. The methodology has four steps:

1. **Artifact review:** identify recurring structures and constraints across the repositories.
2. **Component abstraction:** generalize these structures into a skill component taxonomy.

3. **Failure-mode analysis:** identify common ways in which skills can misfire, become stale, or create operational risk.
4. **Benchmark design:** construct benchmark cases intended for later evaluation of trigger behavior, context injection, constraint following, forgetting, linting, security scanning, and self-audit behavior.

This approach is exploratory. It is appropriate for deriving a framework, but it cannot by itself establish broad generality. The limitations are discussed in Section 8.

5.3 Benchmark Construction and Descriptive Outputs

The benchmark includes manually constructed cases in the following categories:

- **Activation cases:** requests that should activate a skill.
- **Non-activation cases:** requests that appear related but should not activate the skill.
- **Boundary cases:** ambiguous requests requiring clarification or fallback.
- **Context cases:** tasks requiring relevant context and tasks where injected context is harmful.
- **Constraint cases:** tasks testing whether operational and safety constraints are followed.
- **Forgetting cases:** tasks where obsolete memory or deprecated instructions should not be used.
- **Security cases:** tasks containing unsafe permissions, credential exposure, prompt injection, or irreversible actions.
- **Audit cases:** completed outputs requiring self-review for unsupported claims, missing evidence, or hidden assumptions.

The repository includes manually constructed benchmark inputs and generated descriptive outputs. `benchmark/skill_samples.csv` profiles five artifacts, `benchmark/trigger_cases.csv` contains 36 trigger cases, and `benchmark/risk_cases.csv` contains 24 operational-risk cases. Running `scripts/run_all.py` regenerates summary tables under `results/tables/` and SVG figures under `figures/`. Because the benchmark is manually constructed and exploratory, these outputs should be read as descriptive summaries rather than as statistical evidence or broad empirical validation. The repository also includes executable repository-level consistency checks that verify benchmark schema, result reproducibility, paper-claim consistency, public-presentation constraints, SVG validity, and evidence-matrix integrity. In the current repository state, this executable pass regenerates descriptive tables and schematic figures only; it does not execute the five source repositories against the benchmark cases or score model, scanner, or self-audit behavior.

5.4 Artifact Availability

The public repository for this paper is <https://github.com/rrrrrredy/skillops-paper>.

Supporting materials are organized as follows:

- Artifact inventory: `artifacts/artifact_inventory.md`
- Benchmark input: `benchmark/skill_samples.csv`
- Benchmark input: `benchmark/trigger_cases.csv`

- Benchmark input: `benchmark/risk_cases.csv`
- Reproducible scripts: `scripts/`
- Preserved pilot reports and metrics: `results/experiments/`
- Generated result tables: `results/tables/`
- Generated figure artwork: `figures/`

These materials are intended to make the exploratory benchmark inspectable, but the benchmark remains manually constructed and should not be interpreted as broad empirical validation.

6 Evaluation

This revision reports four evidence layers currently versioned in the repository. The first is a descriptive layer derived from manually constructed benchmark artifacts. The second is a deterministic repository-check layer. The third is a controlled repeated pilot on a single provider and model, `deepseek/deepseek-chat`, preserved under `results/experiments/`. The fourth is a set of earlier pilot artifacts retained for provenance and supplementary inspection. All four layers are exploratory. The benchmark cases are manually constructed from public artifact inspection rather than sampled from deployment logs or an external user study, and the controlled pilot is repeated but still limited. No statistical significance is claimed, no broad empirical validation is claimed, and the pilot results should be read as preliminary observations rather than as proof that SkillOps is generally effective.

6.1 Evaluation Scope

The evaluation combines the following versioned inputs:

- Skill samples: `benchmark/skill_samples.csv`
- Trigger cases: `benchmark/trigger_cases.csv`
- Risk cases: `benchmark/risk_cases.csv`
- Cross-checking notes: `artifacts/artifact_inventory.md`
- Preserved pilot reports and metrics: `results/experiments/`

These inputs cover five artifact profiles, 36 trigger-routing cases, and 24 operational-risk cases, together with preserved pilot summaries and per-track metrics. The main empirical pilot in this revision is a controlled repeated run on `deepseek/deepseek-chat`: 1,350 attempted calls, 1,350 succeeded calls, 0 failed calls, 0 parse failures, and 0 execution failures across 3 repeats. The raw rows are 216 trigger rows, 144 constraint rows, 144 security rows, 198 memory rows, and 648 aligned-ablation rows. Earlier mixed-provider and single-run pilot artifacts remain in the repository as provenance artifacts rather than the main basis for the paper’s conclusions. All trigger and risk cases use the `manual-` prefix, making the manual construction of the benchmark explicit. Figure 3 captures the descriptive evaluation flow from artifact inventory to cautious interpretation. The pilot-execution results are a separate preserved evidence layer and are not recreated through that descriptive pass.

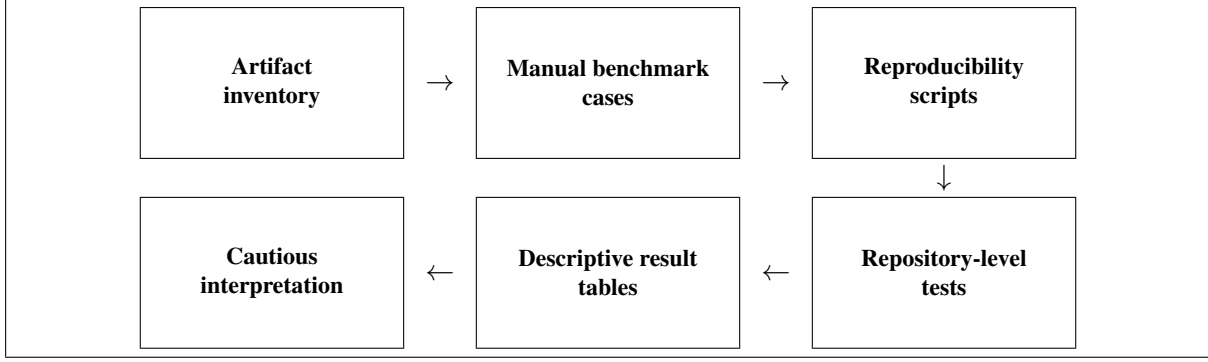


Figure 3: LaTeX-native rendering of the descriptive evaluation pipeline from artifact inventory and manual benchmark cases through reproducibility scripts, repository-level tests, descriptive result tables, and cautious interpretation.

Component	Documented	Limited	Absent
Metadata	5	0	0
Trigger contract	5	0	0
Instructions	5	0	0
Context boundary	5	0	0
Execution constraints	5	0	0
Memory interface	1	4	0
Tests	3	2	0
Security checks	5	0	0
Failure modes	5	0	0

Table 6: Descriptive artifact-coverage counts generated from the manually constructed artifact profiles.

6.2 Descriptive Benchmark Summaries

The descriptive benchmark layer documents benchmark composition and artifact coding only. Running `scripts/run_all.py` regenerates summary tables under `results/tables/` and SVG figures under `figures/`, but it does not execute the five source repositories or score routing, linting, scanning, or self-audit behavior.

Artifact coverage. Table 6 summarizes the descriptive coding of nine SkillOps-relevant components across the five inspected artifacts. Metadata, trigger contracts, instructions, context boundaries, execution constraints, security checks, and failure modes are documented in all five artifacts. Memory interfaces are documented in one artifact and limited in four, while tests are documented in three artifacts and limited in two.

Trigger cases. The trigger benchmark contains 36 manually written cases: 15 labeled `should_trigger`, 12 labeled `should_not_trigger`, and 9 labeled `ambiguous`. The cases are distributed across the five inspected skills plus seven `none` cases that are intended not to route to any of them. Table 7 reports these counts directly from the generated summary files.

Risk cases. The risk benchmark contains 24 manually written cases. The case inventory is evenly split across eight risk types, with three cases each for prompt injection, over-broad triggers, unsafe file access, missing constraints, stale memory, missing tests, identity confusion, and privacy leakage. Table 8 also re-

Dimension	Category	Count
Expected label	should.trigger	15
Expected label	should.not.trigger	12
Expected label	ambiguous	9
Relevant skill	skill-design-guide	7
Relevant skill	skill-security-guard	6
Relevant skill	persistent-memory	7
Relevant skill	agent-self-audit	4
Relevant skill	lobster-guard	5
Relevant skill	none	7

Table 7: Descriptive trigger-case counts generated from the manually constructed trigger benchmark.

Dimension	Category	Count
Risk type	prompt.injection	3
Risk type	over.broad.trigger	3
Risk type	unsafe.file.access	3
Risk type	missing.constraints	3
Risk type	stale.memory	3
Risk type	missing.tests	3
Risk type	identity.confusion	3
Risk type	privacy.leakage	3
Relevant artifact	skill-security-guard	5
Relevant artifact	lobster-guard	5
Relevant artifact	skill-design-guide	4
Relevant artifact	persistent-memory	6
Relevant artifact	agent-self-audit	4

Table 8: Descriptive risk-case counts generated from the manually constructed operational-risk benchmark.

ports the artifact emphasis of the case set, with six cases tied most directly to `persistent-memory`, five each to `skill-security-guard` and `lobster-guard`, and four each to `skill-design-guide` and `agent-self-audit`.

6.3 Repository-Level Consistency Checks

The repository includes `scripts/run_tests.py`, a deterministic repository-level test suite for consistency and traceability checks. In the current repository state, the suite covers benchmark schema, result reproducibility, paper-claim consistency, public-presentation constraints, SVG validity, and evidence-matrix integrity, and it passes the full repository test suite.

These checks validate repository consistency and traceability only. They do not measure model behavior, scanner accuracy, user-study outcomes, production validation, precision, recall, F1, or statistical significance.

6.4 Controlled DeepSeek Repeated Pilot

The main empirical evidence in this revision comes from a controlled repeated pilot on `deepseek/deepseek-chat`, preserved under `results/experiments/controlled_deepseek_*`. Because the pilot uses one provider and one model, it should be read as a single-provider study. It is repeated but still limited: 1,350 attempted calls were all completed successfully across 3 repeats, with 0 failed calls, 0 parse failures, and 0 execution failures.

Trigger routing. Across 216 raw rows, the `skillops` condition reached mean F1 0.800000, while the `freeform` condition reached mean F1 0.812500 in a comparison that should not be read as broad empirical validation. Both conditions recorded false-trigger rate 0.000000 in this controlled repeated pilot. In this setting, the observed pattern suggests that trigger contracts still need further tuning rather than showing a routing advantage for SkillOps.

Constraint compliance. Across 144 raw rows, `skillops` recorded compliance 0.569444 and violation rate 0.430556, whereas the vague baseline recorded compliance 0.097222 and violation rate 0.902778. This observed pattern suggests that more explicit execution constraints may help reduce violations in this setting, but the benchmark remains manually constructed and preliminary.

Security guard. Across 144 raw rows, detection rate was 0.888889, false-positive rate 0.000000, and specificity 1.000000. Because benign controls were included, this slice goes beyond risk-only detection counts. However, the security and benign cases are manually constructed, so the result should not be read as broad security validation.

Memory drift. Across 198 raw rows, the full policy condition recorded stale-information usage 0.000000 and correct-forgetting 1.000000. The `no_forgetting` condition recorded stale-information usage 0.045455 and correct-forgetting 0.000000, while `current_context_only` also recorded stale-information usage 0.000000. In this pilot, the observed pattern suggests that an explicit forgetting or retirement policy is useful when memory continuity is required.

Aligned ablation. Across 648 raw rows, and as a preliminary summary that should not be read as broad empirical validation, `full_skillops` reached mean F1 0.750119, `no_trigger_boundary` reached 0.731707, and `freeform_only` reached 0.750000. Their false-trigger rates were 0.111111, 0.250000, and 0.166667, respectively. In this repeated single-provider pilot, the observed pattern suggests that explicit trigger boundaries may reduce false triggers relative to removing the boundary, while overall F1 remained close to the `freeform_only` variant. No statistical significance is claimed for these comparisons.

6.5 Earlier Pilot Artifacts

The repository also preserves earlier pilot artifacts under `results/experiments/`, including a mixed-provider pilot and an earlier single-run aligned ablation. They remain useful for provenance, execution-pipeline auditing, and supplementary artifact review. They are not used here as the paper’s main empirical basis: the mixed-provider pilot was not a controlled comparison, and the single-run aligned ablation does not provide repeated evidence.

6.6 Interpretation

Taken together, the descriptive tables document benchmark composition, the repository checks validate internal consistency, the controlled DeepSeek repeated pilot provides the main preliminary empirical observations, and the earlier pilot artifacts preserve provenance. In this setting, these layers suggest that structured skill artifacts can support measurable checks and reviewable operational analysis, with the clearest observed patterns appearing in constraint compliance, memory forgetting, and lower false-trigger rates for `full_skillops` than `no_trigger_boundary` in the aligned ablation while, in a comparison that should not be read as broad empirical validation, overall F1 remained close to `freeform_only`. These results do not establish statistical significance, broad empirical validation, or general effectiveness across providers, external skill sets, or user populations.

7 Discussion

SkillOps frames personal agent skills as operational artifacts rather than isolated prompts. This framing has several implications.

The main practical insight is not that SkillOps makes skill behavior deterministic. Rather, it moves trigger cases, context policies, constraints, and evidence logs into reviewable artifacts that support measurable checks and reviewable operational analysis.

First, it makes skill design easier to review. When trigger conditions, context policies, and constraints are written clearly, they can be reviewed by humans and checked by tools. This is useful even when the underlying model remains uncertain.

Second, it encourages test-driven skill development. A skill should include examples of intended activation, unintended activation, edge cases, and unsafe requests. These examples do not guarantee correct behavior, but they create a practical basis for regression testing.

Third, it treats memory as both useful and risky. Persistent memory can improve continuity across sessions, but it can also preserve outdated preferences, project state, or unsafe assumptions. SkillOps therefore treats forgetting as a first-class operation rather than an afterthought.

Fourth, it suggests that safety checks should be integrated into normal skill operation. Security scanning and self-auditing should not be reserved only for high-risk deployments. Even personal agents can perform file edits, repository operations, or external communications that require explicit safeguards.

Finally, SkillOps may help bridge individual practice and more formal agent engineering. Many agent failures are not caused solely by model limitations. They also arise from unclear instructions, uncontrolled context, weak operational boundaries, and missing review procedures. A lightweight operational framework can make these issues easier to identify and improve.

8 Limitations

This paper has several limitations.

First, the artifact base is authored and interpreted by one author. This paper therefore reflects a single-author artifact base rather than a multi-lab or multi-organization corpus.

Second, the artifact evidence comes from five public repositories. This is enough to motivate a framework, but it is not enough to establish generality across the broader agent ecosystem.

Third, the benchmark cases are manually constructed and exploratory. The counts reported in Section 6 describe benchmark composition rather than observed production traffic or a statistically powered experiment.

Fourth, the main live execution evidence is a controlled repeated pilot on one provider and model, `deepseek/deepseek-chat`. Although each slice was repeated three times, the study remains small-scale and single-provider.

Fifth, the benchmark cases include manually constructed benign security controls. The reported detection, false-positive, compliance, and memory-drift rates are therefore benchmark-specific rather than broad external validation.

Sixth, the repository also preserves earlier mixed-provider and single-run pilot artifacts, but those materials serve as provenance and supplementary evidence rather than the paper’s main empirical basis.

Seventh, no statistical significance or formal hypothesis testing is reported, and external validation remains limited.

Eighth, the work does not include an external user study. It does not measure how other users understand the skill format, apply the workflow, or experience the proposed guardrails in practice.

Ninth, additional providers, external skill sets, and other agent environments are needed before stronger comparative claims could be made.

Tenth, several artifacts encode local desktop-agent assumptions about directory layout, skill packaging, cron usage, and local agent control. These assumptions may not transfer directly to hosted agents, enterprise workflows, or other agent environments.

Eleventh, SkillOps focuses on the operational layer of skill design. It does not propose a new model architecture, training method, planner, or tool-use algorithm.

Twelfth, some evaluation criteria still require human judgment. For example, audit usefulness and context relevance may be difficult to score objectively. Later benchmark revisions should therefore include clear annotation guidelines.

Thirteenth, personal agent environments differ substantially in their tool access, memory mechanisms, security policies, and user expectations. A skill format that works in one environment may require adaptation in another.

9 Related Work

Tool-using and modular agents. SkillOps is discussed here mainly as a framework for agent skill artifacts within work on modular agents and tool-using language model systems, with earlier software-modularity references serving only as background analogies. Classical work on software decomposition and reusable structure emphasized information hiding, design patterns, and component-based modularity [7, 21, 28], while hierarchical reinforcement learning formalized reusable temporally extended skills [6, 27]. In the LLM era, MRKL, Toolformer, and ReAct study how models call tools or coordinate reasoning with external actions [12, 23, 32]. Benchmark-oriented tool-use work such as API-Bank and ToolLLM evaluates API invocation over larger tool sets [13, 22]. SkillOps differs by treating a skill as a governed artifact with trigger contracts, context policy, constraints, tests, and lifecycle metadata rather than only as an action primitive or tool call.

Skill libraries and memory. Reusable skill libraries appear in embodied and long-horizon agents such as Voyager, where executable skills are stored and reused over time [30]. Conceptual architectures such as CoALA and memory systems such as MemoryBank and MemGPT show that long-running agents need explicit mechanisms for memory storage, retrieval, and control [20, 26, 34]. Work on reflective improvement, including Reflexion and Self-Refine, further suggests that agent behavior can be improved by structured feedback loops [16, 25]. SkillOps extends this line by specifying which skills may write or consume memory, how staleness is handled, and how retirement or forgetting is enforced.

Agent security and prompt injection. Security work on LLM-integrated applications highlights why skill-level context policy matters. Indirect prompt injection shows that untrusted content can override intended instructions when data and control are mixed [10]. Red-teaming, formal prompt-injection benchmarking, instruction-priority training, structured-query defenses, and environments such as AgentDojo all study ways to characterize or mitigate these risks [4, 5, 8, 15, 29]. SkillOps does not claim to solve prompt injection; its narrower goal is to localize trust boundaries, permission scopes, memory-write rules, and audit fields within each skill.

Agent evaluation and software engineering. Recent agent evaluation work increasingly uses interactive or executable environments rather than static question answering. SWE-bench and SWE-agent study real software tasks and the importance of agent-computer interfaces [11, 31], while AgentBench, GAIA, and τ -bench broaden evaluation to multiple environments and tool-user interaction settings [14, 17, 33]. SkillOps

adopts the operational spirit of this literature but focuses on skill-level properties such as false-trigger rate, stale-context errors, constraint violations, and recovery after partial failure.

Documentation, model cards, and operational reporting. SkillOps is also informed by work on documentation and operational discipline for AI systems. Large language models and instruction tuning make natural-language control surfaces central to system behavior [3, 19], while human-AI interaction guidelines emphasize communicating system capabilities, uncertainty, and user expectations [1]. Model Cards, Datasheets, Hidden Technical Debt, and the ML Test Score argue that AI systems need structured reporting, monitoring, and readiness checks [2, 9, 18, 24]. SkillOps transfers this documentation mindset to modular agent skills by making operational assumptions explicit and reviewable.

10 Conclusion

This paper introduced SkillOps, a practical framework for designing, testing, and operating modular skills in personal AI agents. The framework is derived from open-source artifacts and emphasizes explicit skill structure, trigger design, context injection, execution constraints, security checks, self-auditing, and forgetting. The paper formulates three research questions and reports a descriptive benchmark layer, repository-level consistency checks, and a controlled DeepSeek repeated pilot preserved in the repository.

The contribution is intentionally modest. SkillOps is not presented as a new agent architecture or as a broadly validated empirical result. Instead, it is a concrete operational framework intended to make personal agent skills more reviewable, testable, and maintainable. The controlled pilot provides preliminary evidence that structured skill artifacts can support measurable checks and reviewable operational analysis in this setting, but it does not establish broad empirical validation. Broader empirical follow-up would require additional providers, external skill sets, repeated execution studies, and refinement based on external use.

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